

Mohsen Taheri Shalmani

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EDUCATION

(2020 – 2024) Ph.D. in Information Technology, Mathematics and Physics, University of Stavanger.

(2018 – 2020) MSc in Mathematics and Physics, University of Stavanger.

(2007 – 2009) MSc in Applied Mathematics, Azad University, Science and Research Branch.

(2002 – 2006) BSc in Mathematics, Azad University, Tehran Central Branch.

WORK EXPERIENCE

(2025 – Pres.) AI researcher in [EAST Project](#), Kristiania University of Applied Sciences, Oslo.

(2025 – 2026) Lecturer at the University of Stavanger (20% position).

(2020 – 2024) Doctoral researcher and a member of [BMDLab](#), University of Stavanger.

(2015 – 2018) Math and computer teacher, Mojtama Fani, Tehran.

(2010 – 2015) Bank clerk and IT consultant, Karafarin Bank, Tehran.

(2009 – 2010) Freelance math and computer teacher, Tehran.

(2006 – 2007) Math teacher, Sadr High School, Tehran.

SKILLS

Statistical Modeling and Simulation | R, Python, and MATLAB programming | Mathematical Modeling | AI and ML | Deep Learning | Image Processing and LLM via TensorFlow and Transformers | Big Data | PySpark | Data Mining | Differential Equations | Numerical Analysis | Time Series | Signal Processing | Object Modeling | 3D Slicer | Functional Data Analysis | Language Processing | Computational Geometry | Optimization | Neuroimaging | Object-Oriented Programming | Java, Kotlin, C++, C# | System Testing | Containerization | Docker | Database Validation | Web Applications | React | JavaScript | Power BI | Excel | Cloud Computing | Database Programming | Azure | SQL |

GitHub: <https://github.com/MohsenTaheriShalmani>

Open-Source Contributions

- [EvoMaster](#) – Contributing to advancing the EvoMaster open-source automated testing tool by equipping it with AI technologies to enhance API testing, improving reliability, security, and automated generation of test cases for web and enterprise systems.
- [Elliptical Tubes](#) – Development of the [ETRep R package](#) and [Elliptical Tube Interactive App](#) designed for statistical shape analysis of tubular structures, enabling object modeling, simulation, and mean shape computation of elliptical tubes for applications in biomedical and geometric data analysis.
- [SlicerSALT](#) – Contributor in developing hypothesis testing procedures for shape models within the SlicerSALT open-source platform, which provides advanced tools for statistical shape analysis, visualization, and morphometric measurements for biomedical research.

LANGUAGES

English (Fluent) | Norwegian (Intermediate) | Persian (Native).

SUPERVISION

Co-supervised master's thesis: “*Statistical Shape Analysis of Caudate Nucleus in Parkinson's Disease*” (2025) by Thomas Eikemo Tjensvold, University of Stavanger. <https://hdl.handle.net/11250/3209793>

PUBLICATIONS

- Taheri Shalmani, M., Arcuri, A., Galeotti, J.P., Golmohammadi, A. “Online Machine Learning Prediction of Failed Calls to Enhance REST API Fuzzing”. (2026). Accepted for the 19th International Conference on the Quality of Information and Communications Technology (QUATIC 2026).
- Castagna, F., Galeotti, J.P., Golmohammadi, A., Taheri Shalmani, M., Arcuri, A. “An Empirical Evaluation of Search-Based Algorithms for White-Box Testing of REST APIs”. (2026). Accepted for the International Colloquium on Theoretical Aspects of Computing (ICTAC 2026)
- Arcuri, A., Zhang, M., Seran, S., Galeotti, J.P., Golmohammadi, A., Garrett, P., Sahin, O., Castagna, F., Kertusha, Taheri Shalmani, M., I. and Susilo, F.F. (2026). “EvoMaster at REST League 2026 Tool Competition”. In 19th Search-Based and Fuzz Testing Workshop (SBFT’26).
<https://doi.org/10.1145/3786155.3795705>
- Pizer, S. M., Liu, Z., Zhao, J., Tapp-Hughes, N., Damon, J., Zhang, M., Marron, J. S., Taheri Shalmani, M., & Vicory, J. (2025). “Interior object geometry via fitted frames”. Journal of Mathematical Imaging and Vision. <https://doi.org/10.1007/s10851-025-01266-6>
- Taheri Shalmani, Mohsen (2025) “Deriving Spines of Tubular Objects via Deep Learning. (Preprint)
<https://doi.org/10.13140/RG.2.2.14238.45126>
- Pizer, S. M., Marron, J., Taheri, M., & Vicory, J. “Object Correspondence for Shape Statistics”. (2024) Interactions of Statistics and Geometry (ISAG) II.
- Taheri Shalmani, Mohsen, Stephen M. Pizer, and Jörn Schulz. (2025) [R package] ETRep “Analysis of Elliptical Tubes Under the Relative Curvature Condition”. The Comprehensive R Archive Network (CRAN). <https://doi.org/10.32614/CRAN.package.ETRep>
- Taheri Shalmani, Mohsen. “Shape Statistics via Skeletal Structures.” (2024) Ph.D. Thesis, University of Stavanger, Norway. <https://doi.org/10.13140/RG.2.2.34500.23685>
- Taheri Shalmani, M., Pizer, S. M., & Schulz, J. (2024). “The Mean Shape under the Relative Curvature Condition”. Journal of Computational and Graphical Statistics.
<https://doi.org/10.1080/10618600.2025.2535600>
- Taheri, Mohsen, Stephen M. Pizer, and Jörn Schulz. (2023) “Fitting the Discrete Swept Skeletal Representation to Slabular Objects.” (Under review) <https://doi.org/10.48550/arXiv.2409.04079>
- Taheri, Mohsen, and Jörn Schulz. (2022) “Statistical Analysis of Locally Parameterized Shapes.” Journal of Computational and Graphical Statistics 32, no. 2: 658-670.
<https://www.tandfonline.com/doi/full/10.1080/10618600.2022.2116445>
- Liu, Z., Schulz, J., Taheri, M., Styner, M., Damon, J., Pizer, S., & Marron, J. S. (2023). “Analysis of joint shape variation from multi-object complexes”. Journal of Mathematical Imaging and Vision, 65(3).
<https://link.springer.com/article/10.1007/s10851-022-01136-5>
- Pizer, Stephen M., J. S. Marron, James N. Damon, Jared Vicory, Akash Krishna, Zhiyuan Liu, and Mohsen Taheri. (2022) “Skeletons, object shape, statistics.” Frontiers in Computer Science 4: 842637.
<https://www.frontiersin.org/articles/10.3389/fcomp.2022.842637/full>
- Taheri Shalmani, Mohsen. (2020) “Statistical shape analysis of brain structures.” Master's thesis, University of Stavanger. <https://doi.org/10.13140/RG.2.2.12270.42567>

REFERENCES

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| • Prof. Jörn Schulz (UiS, Norway) | Email: jorn.schulz@uis.no |
| • Prof. Jan Terje Kvaløy (UiS, Norway) | Email: jan.t.kvaloy@uis.no |
| • Prof. Andrea Arcuri (Kristiania University, Norway) | Email: andrea.arcuri@kristiania.no |
| • Prof. Stephen M. Pizer (UNC, USA) | Email: smp@cs.unc.edu |